

Diagnosing the Mind: A Machine Learning Approach to Early Detection of Alzheimer's Disease

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I. ABSTRACT

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder that significantly affects memory, cognition, and behavior. Early diagnosis is crucial, as it enables timely medical intervention and lifestyle adjustments that can help slow the disease's progression. However, traditional diagnostic methods like MRI scans and cognitive tests suffer from limitations such as subjectivity, high costs, and inconsistency. This study aims to develop a machine learning-based classification system that predicts Alzheimer's Disease stages using diverse patient-related data including demographics, medical history, clinical indicators, and cognitive scores. Models such as Random Forest and Decision Tree are evaluated based on their precision, recall, and feature importance to identify the most influential predictors. The project also seeks to differentiate between stages of dementia, offering a pathway toward scalable, efficient, and early diagnosis of Alzheimer's Disease.

II. INTRODUCTION

Alzheimer's Disease (AD) is one of the most common causes of dementia among older adults. It involves the gradual deterioration of brain cells due to abnormal protein build-up, leading to cognitive decline and memory loss. Despite being incurable, early detection of AD can help mitigate its progression through timely treatment and support strategies. In the era of digital health, machine learning (ML) presents a promising approach for early and accurate detection of Alzheimer's Disease by analyzing large volumes of multi-dimensional patient data.

Traditional methods of diagnosing Alzheimer's Disease rely heavily on subjective assessments, such as neuropsychological evaluations, or expensive imaging techniques like MRI. These methods often fail to detect the disease in its early stages or provide inconsistent results. Given the growing aging population and the associated rise in dementia cases, there is a critical need for an automated, scalable, and data-driven diagnostic approach. Machine learning algorithms can uncover hidden

patterns in patient data and offer robust predictions, making them a valuable tool in modern medical diagnostics.

The primary objective of this research is to develop a machine learning-based classification model capable of predicting whether a patient has Alzheimer's Disease and, if so, determining the severity stage. The dataset includes a wide range of features such as age, gender, education level, Mini-Mental State Examination (MMSE) scores, and other clinical and demographic variables. The study focuses on:

- Building classification models to identify Demented vs. Non-Demented individuals.
- Differentiating between Mild and Moderate dementia.
- Evaluating and comparing the performance of Random Forest and Decision Tree classifiers.
- Analyzing feature importance to determine key predictors for AD diagnosis.

This project aims to enhance the reliability, accessibility, and efficiency of Alzheimer's Disease diagnostics using data-driven approaches.

III. RELATED WORK

"Most illness attack the body, Alzheimer's destroy the mind."- this common disease has taken an uncountable number of peoples' precious memories that they have collected throughout their lives. Although there is no specific cure for this disease, still early detection of it can slow down the progression in order to give the patients a better life. A number of research studies have been done to identify Alzheimer's disease at an early stage. Through our research we aim to contribute in the society.

The study by Liu et al. (2022) presents a novel 3D convolutional neural network (CNN) model for the early detection of Alzheimer's Disease (AD) using structural MRI scans [2]. Cognitively normal people, people with moderate cognitive impairment (MCI), and those with mild AD dementia are all successfully distinguished by the model. With a high accuracy (AUC of 85.12 on external validation), it performs better than conventional models that depend on

manually derived characteristics like brain volume and cortical thickness. The model also showed promise in forecasting the transition from MCI to AD. It is more quicker and requires less human preparation than traditional approaches. Better generalization is provided by the model, which also identifies larger brain areas associated with AD. Although the work shows promise, it might be improved by expanding the dataset, incorporating multimodal data (such as genetic and clinical data), and improving interpretability for clinical applications.

Another paper titled "Machine Learning Techniques for the Diagnosis of Alzheimer's Disease: A Review" by M. Tanveer et al. (2020) provides a comprehensive review of 165 studies from 2005 to 2019, focusing on machine learning (ML) techniques for diagnosing Alzheimer's disease (AD) [5]. In order to categorize AD stages using neuroimaging data such as MRI, PET, and DTI, the goal is to survey and evaluate the efficacy of machine learning (ML) models, such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Deep Learning (DL). The authors point out that machine learning (ML) greatly increases diagnostic accuracy, especially when it comes to differentiating between AD patients, control individuals, and people with mild cognitive impairment (MCI).

Amulya E. R. et al. (2016) conducted research on the classification of brain images for the early detection of Alzheimer's Disease [1]. They have used various methods for accurate and reliable results. Voxel based morphometry (VBM) is used for extracting significant features from brain MRI, Principal Component Analysis (PCA) is used for feature selection. Techniques like PCA, SVM-RFE, and PSO are used to reduce complexity and boost accuracy. A multi-layer neural network with a sigmoid activation function is suitable for identifying non-linear patterns in data. Another effective method, the Extreme Learning Machine (ELM) is used. The authors proposed a novel hybrid model highlighting their accuracy rates, sensitivity, and specificity.

Another research by M Mahyoub et al (2018) demonstrated an approach for predicting Alzheimer's disease at an early stage based on different risk factors using an ensemble of various weak classifiers [3]. The project aimed for an early prediction of the disease prior to any brain imaging to reduce inaccurate diagnosis by general practitioners. The classification is based on 9 risk factors from medical history, lifestyle, and demography. Even though the model initially classified the disease with high specificity and precision, its performance reduced significantly with the ensembled model. The authors concluded that a dataset with more risk factors may improve the overall performance. This project identified a research scope that may help identifying the more risk factors and impact matrix of the risk factors and identifying the disease with better performance.

The paper authored by Y Fu et al (2024) presented a state-

of-art model that uses a multimodal for identifying Alzheimer's disease [4]. The classification is done by analyzing both acoustic signals and content vs context relationship in the extracted text from the speech. The authors of the study extracted the acoustic features using the ImageBind audio encoder, and obtained the semantic features using ELMo from raw speech data. They effectively mapped the images to 6 modalities to include text and speech in the same joint embedding space to achieve a cross-modal semantic alignment. This approach achieved excellent results with 100% recall and 90% accuracy. However, the authors pointed out that the efficiency can be further improved by including selective imaging data and including more speech data.

IV. DATASET INFORMATION

Dataset Name: Alzheimer's Disease Dataset

Source Link: <https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset>

Format: CSV

Description:

The dataset contains health information for 2,149 individuals, both diagnosed and not diagnosed with Alzheimer's disease. It includes demographic details, lifestyle factors, medical history, clinical measurements, cognitive and functional assessments, symptoms, and more. The data is organized in a tabular format with boolean and numeric values. The target variable is `Diagnosis`, which has two unique values:

- 0 – Alzheimer's negative
- 1 – Alzheimer's positive

Risk factors such as demographics of the patient, lifestyle choices and medical history are used to classify the diagnosis of Alzheimer's.

V. METHODOLOGY

A. Initial Exploratory Data Analysis

The dataset contains 2149 data of Alzheimer's patients that includes 35 different features each of which are potential risk factors for the disease.

There's only one categorical feature `DoctorInCharge` which has only one unique value for all of the data points, that is a flag stating that the actual data is hidden due to privacy concerns. The `PatientID` feature has unique values for each of the data points. Therefore, we are not getting any information influencing the prediction.

The following numeric values actually contain categorical data:

- Gender
- Ethnicity
- EducationLevel
- Smoking
- FamilyHistoryAlzheimers
- CardiovascularDisease
- Diabetes
- Depression
- HeadInjury

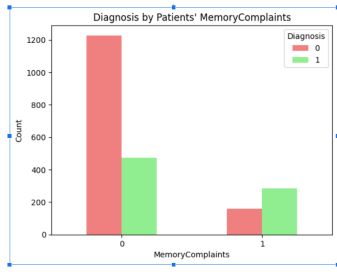


Fig. 1. MemoryComplaints and Diagnosis

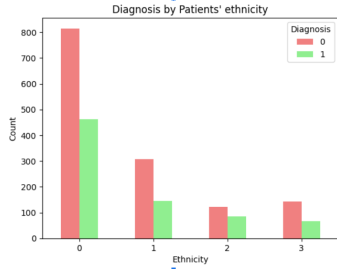


Fig. 2. Ethnicity

- Hypertension
- MemoryComplaints
- BehavioralProblems
- Confusion
- Disorientation
- PersonalityChanges
- DifficultyCompletingTasks
- Forgetfulness
- Diagnosis

The nominal data are also encoded with numerical values which might add bias towards certain values.

However, no null values are found in the data set. The presumed risk factors did not show significant relationships with the diagnosis.

The relationship between *MemoryComplaints* and *Diagnosis*:

The relationship between *Ethnicity* and *Diagnosis*:

The correlation heatmap is generated from the correlation matrix using a threshold of 0.3. The *FunctionalAssessment*, *MemoryComplaints* and *ADL* features are found to have correlations over 0.3 with the target feature *Diagnosis*.

B. Data Preprocessing

The dataset was sourced from Kaggle, titled “Alzheimer’s Disease Dataset” by rabieelkharoua. The dataset includes patient demographics, lifestyle factors, medical history, and Alzheimer’s-related assessments. There are 2149 rows and 35 columns. There are no missing values in any column. The PatientID column was dropped as it was irrelevant. The dataset consists of 22 integer columns, 12 float columns, and 1 categorical column (DoctorInCharge), which is of object type. One-Hot Encoding was applied to the categorical column DoctorInCharge to transform it into binary features.

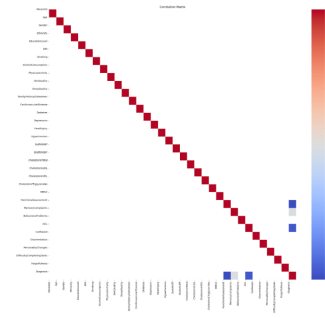


Fig. 3. Figure: Correlation Hitmap

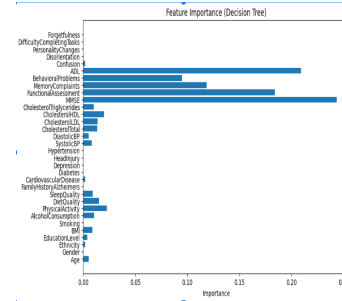


Fig. 4. Figure: Decision Tree

Numerical columns such as Age, BMI, and CholesterolTotal were standardized using StandardScaler for consistent scaling. The column *Diagnosis* likely serves as the target variable for prediction.

C. Models

A baseline model is a simple machine learning model that is used to set a performance benchmark. It helps us to understand whether more complex models are actually improving prediction or not. From the dataset we could see that it is a tabular data, so we choose a Decision Tree Classifier as the baseline because it is easy to interpret and works well with tabular medical data. The Alzheimer’s dataset, which comprises clinical scores, symptom information, and patient demographics, is being used by the model. The MMSE score, which is frequently used to measure cognitive impairment, was one of the most significant variables that the tree automatically chose. The decision tree that resulted had seven leaf nodes and a depth of four, indicating that the model used four decision steps to arrive at a classification.

The baseline model successfully distinguished between those with and without dementia, with an accuracy of almost 81%. The MMSE score was found to be the most significant characteristic, followed by clinical history and age.

Tools: Pandas, NumPy, Seaborn, KaggleDatasetAdapter, sklearn

VI. RESULTS AND EVALUATION

Using a dataset from Alzheimer’s disease, we assessed five machine learning models: Decision Tree, Random Forest,

Model	Accuracy
Decision Tree	0.906977
Random Forest	0.925581
Logistic Regression	0.830233
SVM	0.644186
Ensemble (Voting)	0.927907

Fig. 5. Figure: Accuracy of Models

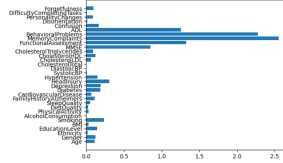


Fig. 6. Figure: Random Forest

Logistic Regression, Support Vector Machine (SVM), and an Ensemble Voting Classifier. Metrics like accuracy, precision, recall, F1-score, and confusion matrix were used to evaluate performance. The Ensemble model outperformed the others in terms of accuracy, proving that combining several models enhances predictive performance by lowering variance and bias. The Decision Tree model fared comparatively poorly, maybe as a result of overfitting, but the Random Forest model performed well because of its strong decision tree averaging. Bar plots for Random Forest and Logistic Regression were used to illustrate the significance of each feature. These plots helped us comprehend the model's decision-making process by highlighting the most significant features that contributed to the forecast. Random Forest revealed which features were most commonly employed in decision splits, whereas Logistic Regression coefficients highlighted the strength and direction of feature influence.

With the majority of predictions falling on the diagonal (correct classifications), the Ensemble model's confusion matrix showed that it made the fewest misclassifications when compared to other models. It was evident from a comparison table of accuracies that the Ensemble model performed better than the others. The Ensemble approach is confirmed to be the most dependable option for this classification assignment by these assessments.

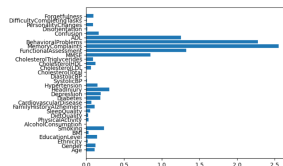


Fig. 7. Figure: Logistic regression

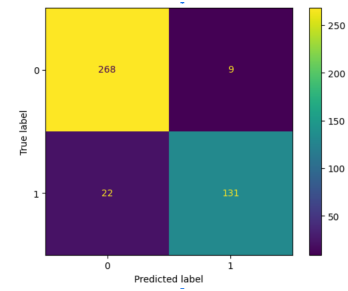


Fig. 8. Figure : Confusion matrix

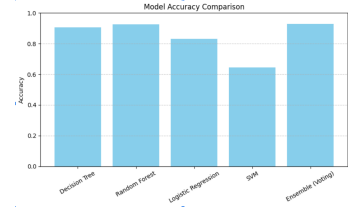


Fig. 9. Figure: Model Accuracy Comparison

VII. DISCUSSION

From a diverse set of data on Alzheimer's patients, we were able to identify some of the risk factors. Most of the features have very little influence on the identification of the disease. However, some of the characteristics contributed to the classification process for the identification of the disease. We have observed the feature importance and feature influences in different models. The most contributing risk factors include the *Cognitive and Functional Assessments* data including *ADL(Activities of Daily Living score)*, *memory complaints*, *functional assessment data*, *MMSE(Mini-Mental State Examination score)*, etc.

In addition to the cognitive and functional evaluation data, some of the lifestyle factors were found to contribute to the classification of the disease. Among lifestyle factors, sleep quality, diet quality, and physical activity were the most significant.

However, none of these risk factors were found significantly influential to the models where feature importance factors were less than 20% for every case. Except for the cognitive, functional, and lifestyle factors, most of the features were very little influential, contributing less than 0.02% in the importance factors for all of the models.

Most of the models classified the disease with accuracy ranging from 80% to more than 90%. Except for the Support Vector Machine which predicted with only 64% accurately. An ensemble of Decision Tree, Random Forest, and Logistic Regression yield improved the performance with 93% accuracy with less false positive and false negative classifications.

VIII. CONCLUSION

The ensemble model was successful in classifying the disease in most of the performance matrices with the risk factor features. However, this prediction may not be taken as

a conclusive diagnosis of the disease as the risk factors were found to have lower influence for the diagnosis. Instead, this prediction may be used as a preliminary diagnosis before going into more complex, time consuming, and expensive diagnosis procedures.

This early prediction process may be further improved by adding more samples to the datasets. More cognitive and function assessment data and lifestyle factors data may be collected to improve the performance as these features were found as the most influential risk factors.

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